Write a Servlet to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.

Aim

To demonstrate the functional differences between the **HTTP GET** and **POST** methods in a Java Servlet by creating an interactive login/feedback form for the "AI-ML Department Webpage."

Procedure

Environment Setup: Ensure Apache Tomcat server is configured in your IDE (Eclipse/IntelliJ).
HTML Creation: Create an `index.html` file containing a form with two buttons: one sending data via GET and another via POST.
Servlet Development: Write a Java class `DepartmentServlet` that extends `HttpServlet`.
Method Implementation: * Override `doGet()` to handle data appended to the URL. Override `doPost()` to handle data sent in the request body.
Mapping: Map the Servlet in `web.xml` or use the `@WebServlet` annotation.
Deployment: Run the project on the server and observe the URL behavior for both methods.

Code

index.html

HTML

```
<!DOCTYPE html>
<html>
<head>
    <title>AI-ML Department Webpage</title>
```

```
</head>
<body>
    <h2>Welcome to the AI-ML Department</h2>
    <p>Testing HTTP Methods for Student Registration:</p>

    <form action="DepartmentServlet" method="GET">
        <input type="text" name="studentName" placeholder="Enter Name
(GET)" required>
        <button type="submit">Submit via GET</button>
    </form>

    <br>

    <form action="DepartmentServlet" method="POST">
        <input type="text" name="studentName" placeholder="Enter Name
(POST)" required>
        <button type="submit">Submit via POST</button>
    </form>
</body>
</html>
```

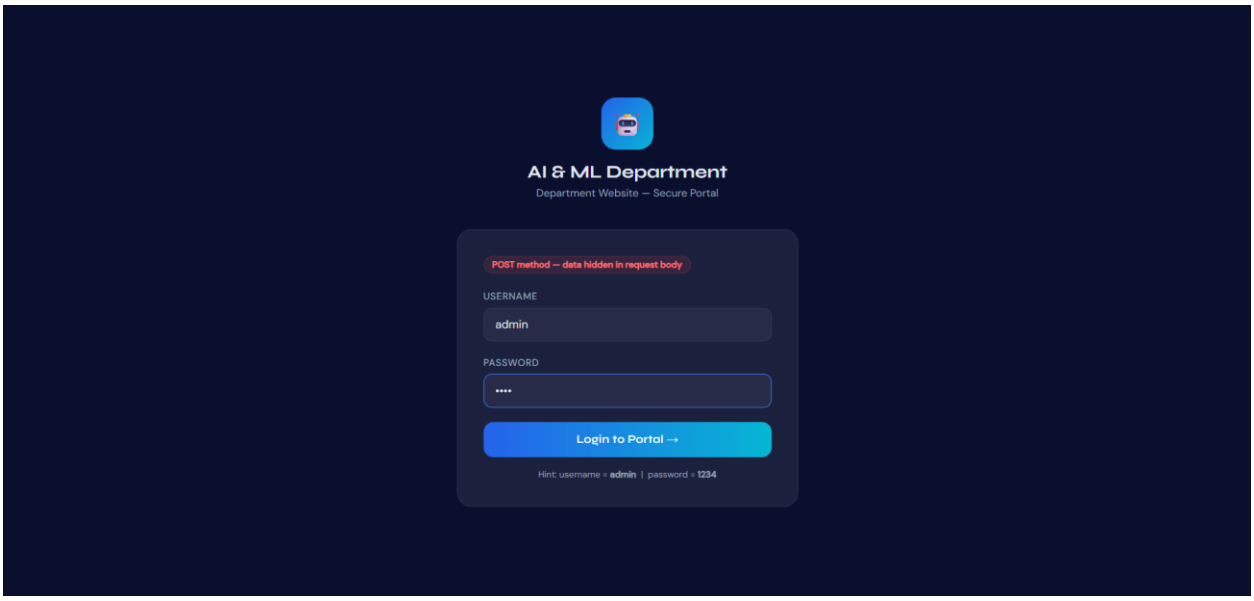
DepartmentServlet.java

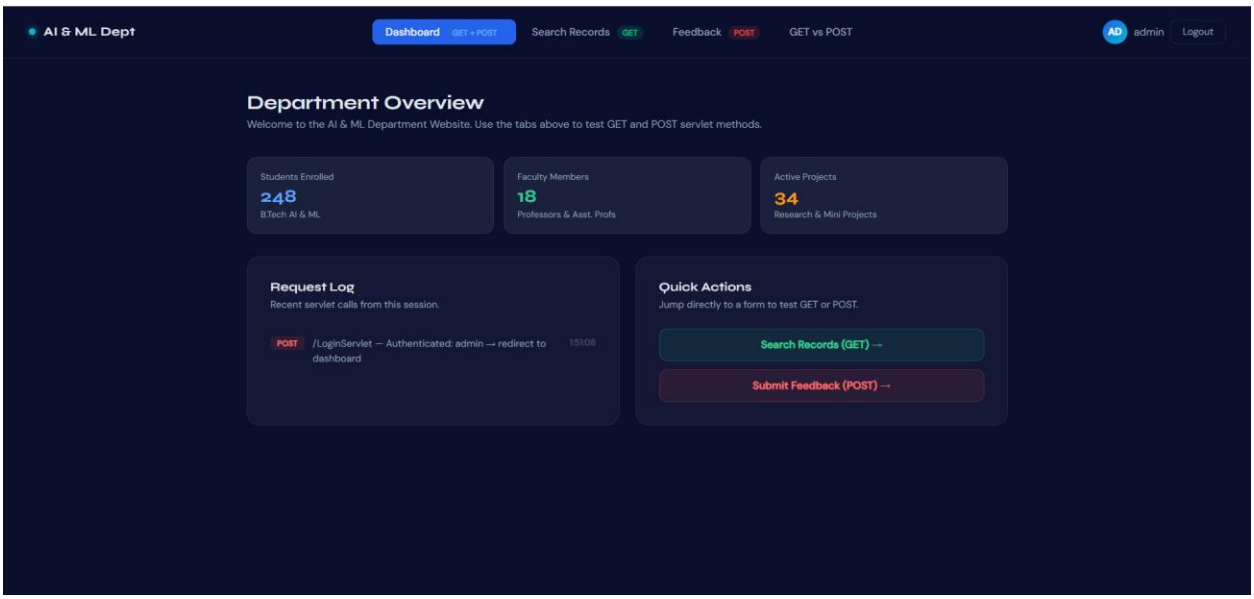
Java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/DepartmentServlet")
public class DepartmentServlet extends HttpServlet {

    // Handles GET requests
    protected void doGet(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String name = request.getParameter("studentName");
```

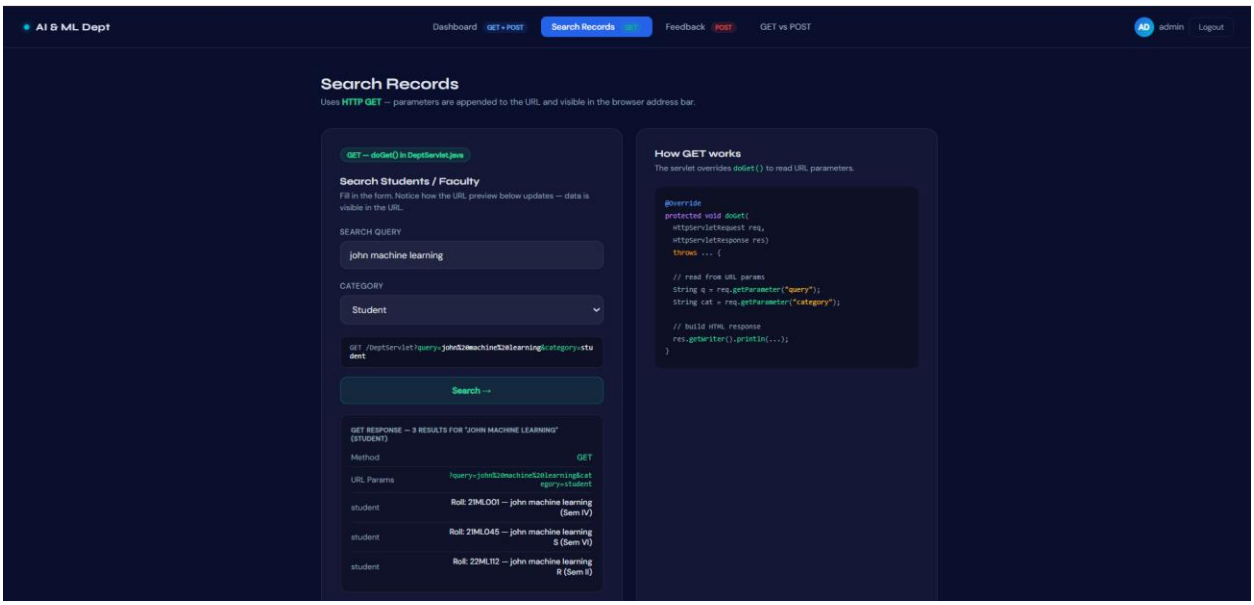
```
        out.println("<html><body>");
        out.println("<h3>AI-ML Department: GET Request
Received</h3>");
        out.println("<p>Data visible in URL: <b>" + name +
"</b></p>");
        out.println("</body></html>");
    }

    // Handles POST requests
    protected void doPost(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String name = request.getParameter("studentName");

        out.println("<html><body>");
        out.println("<h3>AI-ML Department: POST Request
Received</h3>");
        out.println("<p>Data sent securely in body: <b>" + name +
"</b></p>");
        out.println("</body></html>");
    }
}
```

Output

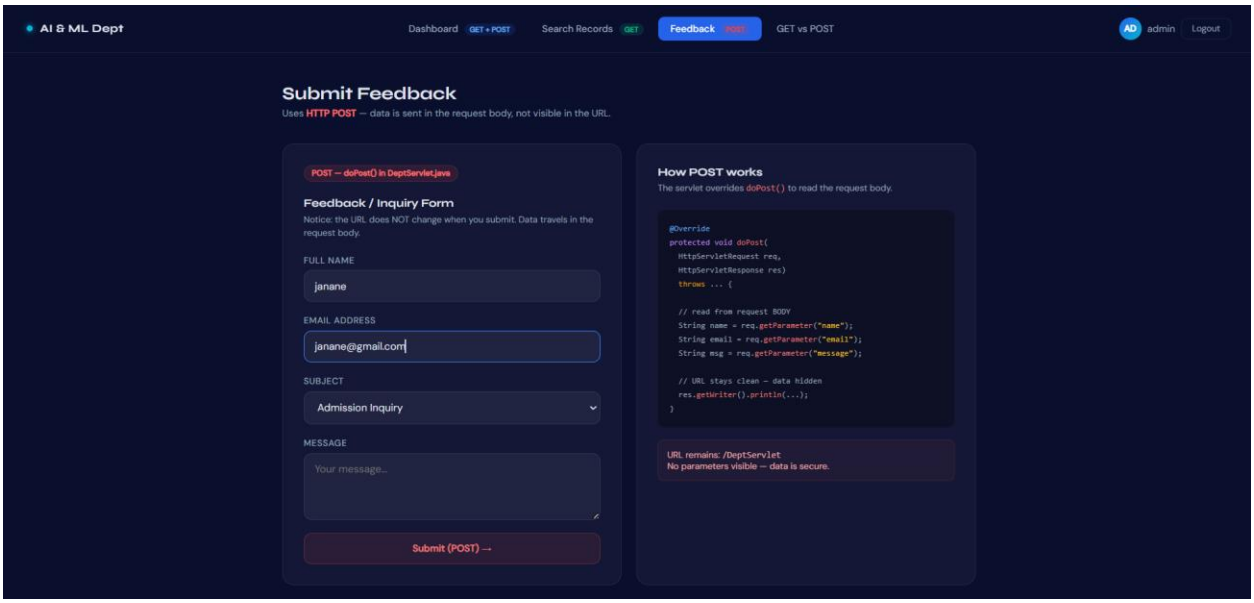

AI & ML Department
Department Website — Secure Portal
POST method — data hidden in request body
USERNAME
admin
PASSWORD
Login to Portal →
Hint: username = admin | password = 1234



GET Form:



POST FORM:



Result :

The experiment successfully demonstrates that for the AI-ML department webpage, **GET** is suitable for fetching data, while **POST** should be used for submitting sensitive student information.